Energy Management Approaches

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Courtesy for some slides: Dr. Rajeev Balasubramonian

Frequency of Various Processors

	Freq (MHz)		Freq (MHz)
Intel's 486	66	TPU v1/v2	700
Intel E5-2697v2 CPU	2700	TPU v3	940
Intel i7-8700K CPU	3700	TPU v4	1050
Brainwave FPGA	500	IPU1	1600
Pascal GPU in Jetson	1300	Intel Atom x5-Z8350	1920
Tesla K40 GPU	745	KNL Phi	1300
V100 GPU	1500	KNM Phi	1500
A100 GPU	1400	MTIA v1 (from Meta)	800
ARM Cortex M-7	<600MHz	MTIA v2 (from Meta)	1350

- CPUs use high frequency, whereas other processors use low frequency.
- We will explain the reason behind this.

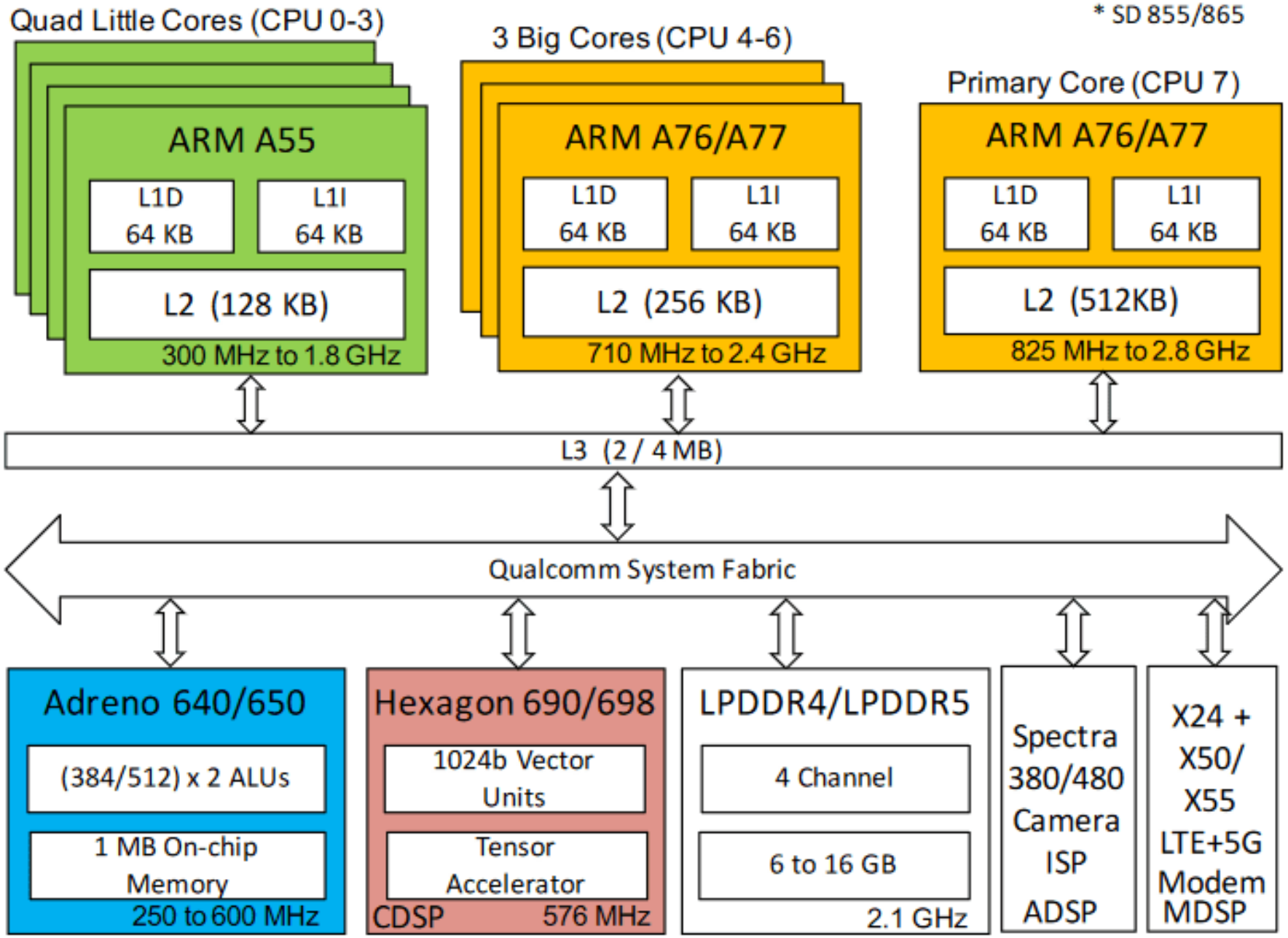

Fig. 1: Snapdragon 855/865 with heterogeneous compute unit.

Power vs. Energy

- Energy is the ultimate metric: it tells us the true "cost" of performing a fixed task

- Power is energy per unit time
- Power poses constraints. We can only work fast enough to max out the power delivery or cooling solution

Example

- Processor A consumes 1.2× the power of processor B, but finishes the task in 30% less time. Which processor takes less energy?
- Solution: Processor A's relative energy is 1.2×0.7 = 0.84
- Processor A is better, assuming that 1.2× power can be supported by the system

	B	A
Power	100 units	120 units
Time	100 units	70 units
Energy	10,000 units	8,400 units

Components of Power

- Power = static power + dynamic power
- Energy = Power × time

  = (DynanmicPower + LeakagePower) × time

Static Power (also called leakage power)

Static power consumption

- $\text{Current}_{\text{static}} \times \text{Voltage}$.
- Scales with number of transistors.
- *Static power* is dissipated by leakage currents that flow even when the device is inactive.

Dynamic Power

- Dynamic energy
  - Transistor switch from $0 \rightarrow 1$ or $1 \rightarrow 0$
  - $\frac{1}{2} \times$ Capacitive load $\times$ Voltage^2

- Dynamic power
  - $\frac{1}{2} \times$ Capacitive load $\times$ Voltage^2 $\times$ Frequency of switching

- Reducing frequency (clock rate) reduces power, not energy

Problem 1

For a processor running at 100% utilization at 100 W, if 20% of the power is attributed to leakage, then what is the total power dissipation when the processor is running at 50% utilization?

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$$\begin{aligned}\text{Total power} &= \text{dynamic power} + \text{leakage power} \\ &= 80\ \text{W} \times 50\% + 20\ \text{W} \\ &= 60\ \text{W}\end{aligned}$$

Problem 2

If processor A consumes 1.4× the power of processor B but finishes the task in 20% less time, which processor would you pick?

(a) If you were constrained by power delivery constraints?

(b) If you were trying to minimize energy per operation?

(c) If you were trying to minimize response times?

Problem 2

If processor A consumes 1.4× the power of processor B but finishes the task in 20% less time, which processor would you pick?

(a) If you were constrained by power delivery constraints?
Processor B

(b) If you were trying to minimize energy per operation?
Energy of processor A is 1.4×0.8 = 1.12 times the energy of processor B. We choose Processor B

(c) If you were trying to minimize response times?
Processor A, as it is faster

Relation Between Frequency and Time

When a processor's frequency is

(a) increased by 30%

(b) decreased by 40%

find the percentage change in execution time.

Answer: (a) New time = 1/130% = 100/130 = 0.769 = 76.9%

Thus, execution time has reduced by 23.1%.

(b) New time = 1/60% = 100/60 = 1.66 = 166.66%

Thus, execution time has increased by 66.66%.

DFS and DVFS

Energy/Power Saving Techniques

Reducing Power and Energy

DFS: Dynamic frequency scaling

- Control knob: reduce frequency
- Result: reduces dynamic power, but increases energy

DVFS: Dynamic voltage and frequency scaling

- Control knob: reduce frequency and voltage
- Result: reduces both dynamic and static power and energy
- Voltage drop leads to slow transistors so frequency of operation is also reduced

Relationship b/w voltage and frequency

Voltage drop leads to slow transistors, so frequency of operation is also reduced.

When you increase frequency, you have to also increase the voltage.

Problem 3

If processor A at 3 GHz consumes 80 W of dynamic power and 20 W of static power and completes a program in 20 sec, then

A. Find energy of the processor.

B. Find energy on scaling frequency down by 20%.

C. Find energy on scaling frequency and voltage down by 20%.

D. Find energy on scaling frequency up by 20%. [Not feasible]

E. Find energy on scaling frequency and voltage up by 20%.

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New dynamic power = 64 W; New static power = 20 W

New execution time = 25 sec (assuming CPU-bound)

Energy = 84 W × 25 sec = 2100 J

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New dynamic power = 64 W; New static power = 20 W

New execution time = 25 sec (assuming CPU-bound)

Energy = 84 W × 25 sec = 2100 J

C. Energy on scaling frequency and voltage down by 20%

New dynamic power = 41 W; New static power = 16 W;

New execution time = 25 sec; Energy = 1425 J

Problem 3

If processor A at 3 GHz consumes 80 W of dynamic power and 20 W of static power and completes a program in 20 sec, then

D. Energy on scaling frequency up by 20%

New execution time = 20/1.2 = 16.7 sec

New dynamic power = 80×1.2 = 96 W; New static power = 20 W

Energy = (96+20) W × 16.7 sec = 1937.2 J

E. Energy on scaling frequency and voltage up by 20%

New dynamic power = 80×1.2^3 W=138.2 W;

New static power = 20×1.2=24 W; New execution time = 16.7 sec

Energy = (138.2+24)×16.7 = 2708.7 J

Summary of Results

	Energy
Original energy	2000 J
Scaling frequency down by 20%	2100 J
Scaling frequency and voltage down by 20%	1425 J
Scaling frequency up by 20%	1937 J
Scaling frequency and voltage up by 20%	2709 J

- For reducing energy consumption, voltage and frequency need to be scaled together (DVFS).
- Increasing voltage/frequency by just 20% increases energy by 35%. Hence, most processors do not use high frequency.
- CPUs use high frequency to reduce execution time.

Example

A particular cell phone has a 6 watt-hour (W-hr) battery and operates at 1.2 V.

When it is in use, the cell phone operates at 300 MHz and the average amount of capacitance in the chip switching at any given time is 10 nF. When in use, it also broadcasts 3 W of power out of its antenna.

When the phone is not in use, the dynamic power drops to almost zero because the signal processing is turned off. But the phone also draws 40 mA of quiescent current whether it is in use or not.

Determine the battery life of the phone

(a) if it is not being used, and

(b) if it is being used continuously.

Solution

The static power is Pstatic = (0.040 A)(1.2 V) = 48 mW.

When not used, this is the only power consumption, so the battery life is (6 Whr)/(0.048 W) = 125 hours.

If the phone is being used,

Pdynamic = $(0.5)(10^{-8}\ \text{F})(1.2\ \text{V})^2(3*10^8\ \text{Hz}) = 2.16$ W.

Together with the static and broadcast power, the total active power is 2.16 W+0.048 W +3W = 5.2W,

So the battery life is 6 W-hr/5.2 W = 1.15 hours.